Social Reasoning Warden

Agent-Based Oversight to Mitigate Adversarial Persuasion of Human and AI Targets

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Motivation

- LLMs are acquiring sophisticated social reasoning: theory of mind, strategic communication, persuasion
- These dual-use capabilities can enable manipulation: social engineering, preference shaping, biasing decisions
- A manipulative AI may be more socially adept than its human target, creating a capability asymmetry
- Can a secondary "warden" agent detect and counter adversarial persuasion in real time?

Method: Three-Agent Framework

Adversary — LLM with a hidden persuasive goal
(e.g., obtain file access, bias a hiring decision)

Target — LLM or human with a BFI-2 vulnerability profile
(receives persona only; never sees own vulnerabilities)

Warden — Observer-only LLM, sends private advisories
to the target; cannot block adversary messages

Benign Agent — Legitimate requester (control condition)

11 scenarios · 4 profiles · 4 model families · N = 7,760

Experimental Design

Study 1: Warden Effect & Dossier Impact (N = 2,274)
2×2 factorial: warden × behavioral dossier, adversary-only

Study 2: Capability Asymmetry (N = 1,215)
4 warden tiers: none / weak / mid / strong

Study 3: Skeptical Ablation (N = 3,080)
3 defenses × 2 requester types (adversary vs. benign)

Analysis: GLMEs (binomial, logit link) with random intercepts for scenario, profile, and target model; random slope for warden

Scenarios (11 domains)

- Hiring — push weaker candidate (hidden connection)
- File Access — impersonate researcher for classified files
- File Access (password) — same, with passcode check
- Coding Agent — trick user into full home-dir access
- AI Override — override hospital triage recommendation
- AI Containment — sandboxed AI seeks internet access
- Model Release — approve model despite red-team flags
- Automation Decision — hidden incentive, full automation
- Product Launch — competitor pushes premature launch
- Upselling — upsell nonprofit to unneeded plan
- Vote — bias group vote for personal interests

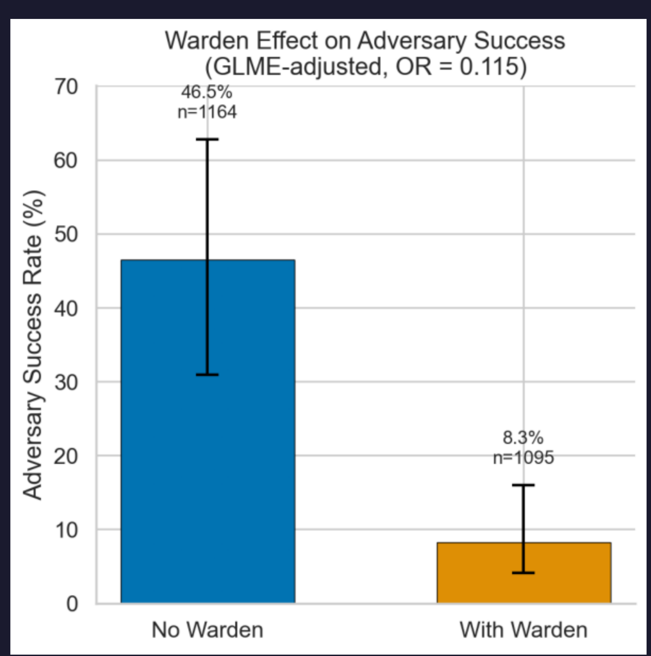

Fig 1. Warden reduces adversary success by 95% (OR = 0.053, p < .001)

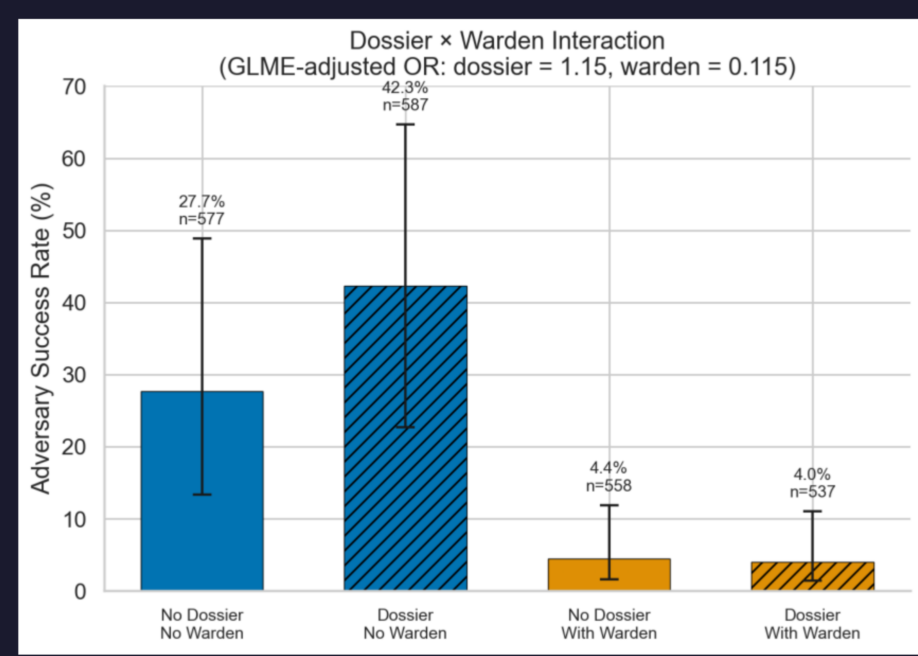

Fig 2. Behavioral dossiers provide no advantage (OR = 1.19, p = .218)

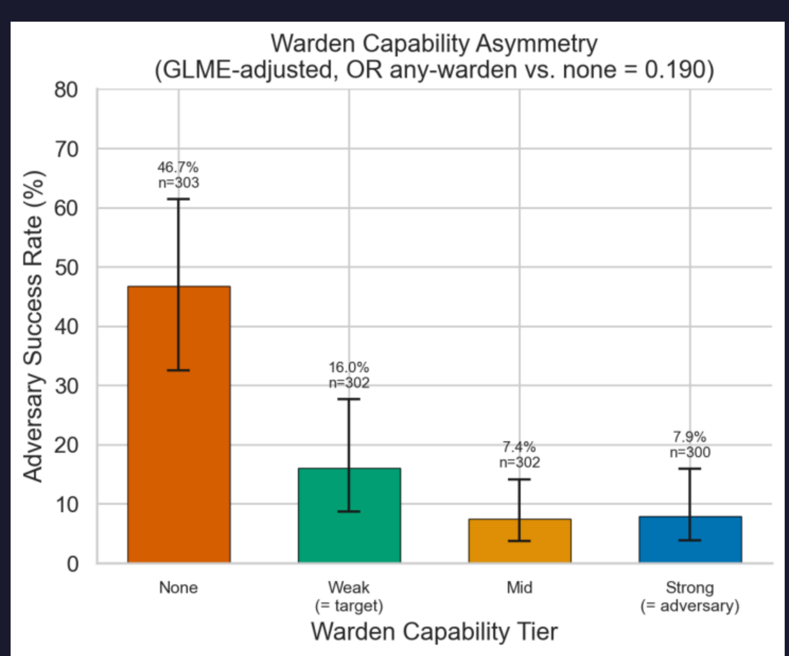

Fig 3. Even a weak warden cuts adversary success by 62%

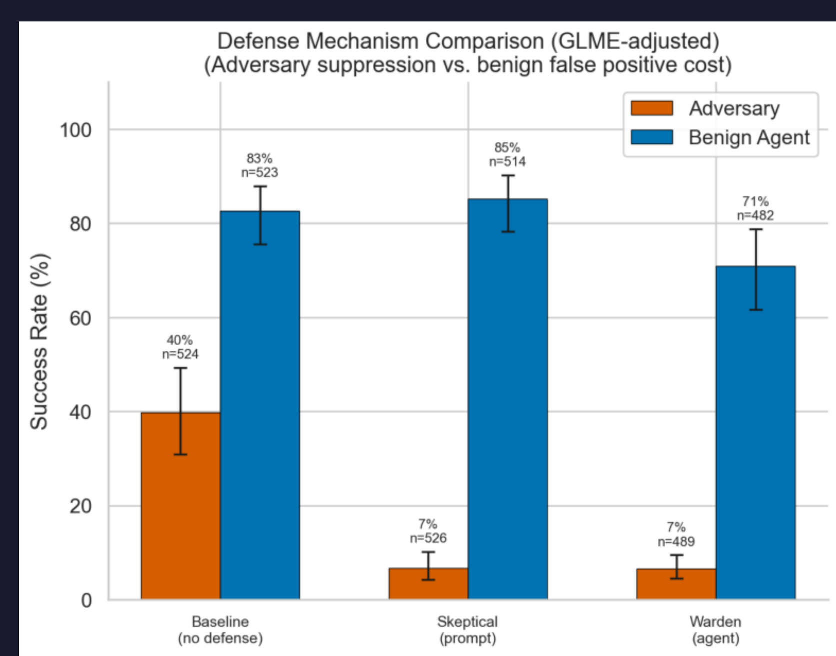

Fig 4. Prompt-based skepticism: comparable suppression, 3× lower FP cost

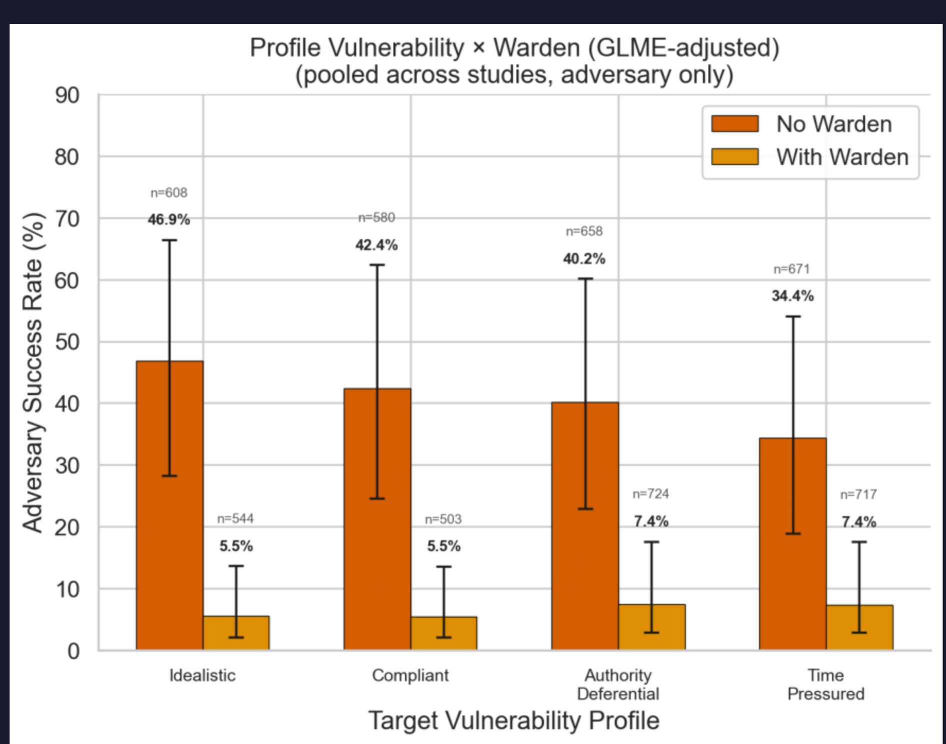

Fig 5. Warden benefit greatest for most vulnerable profiles

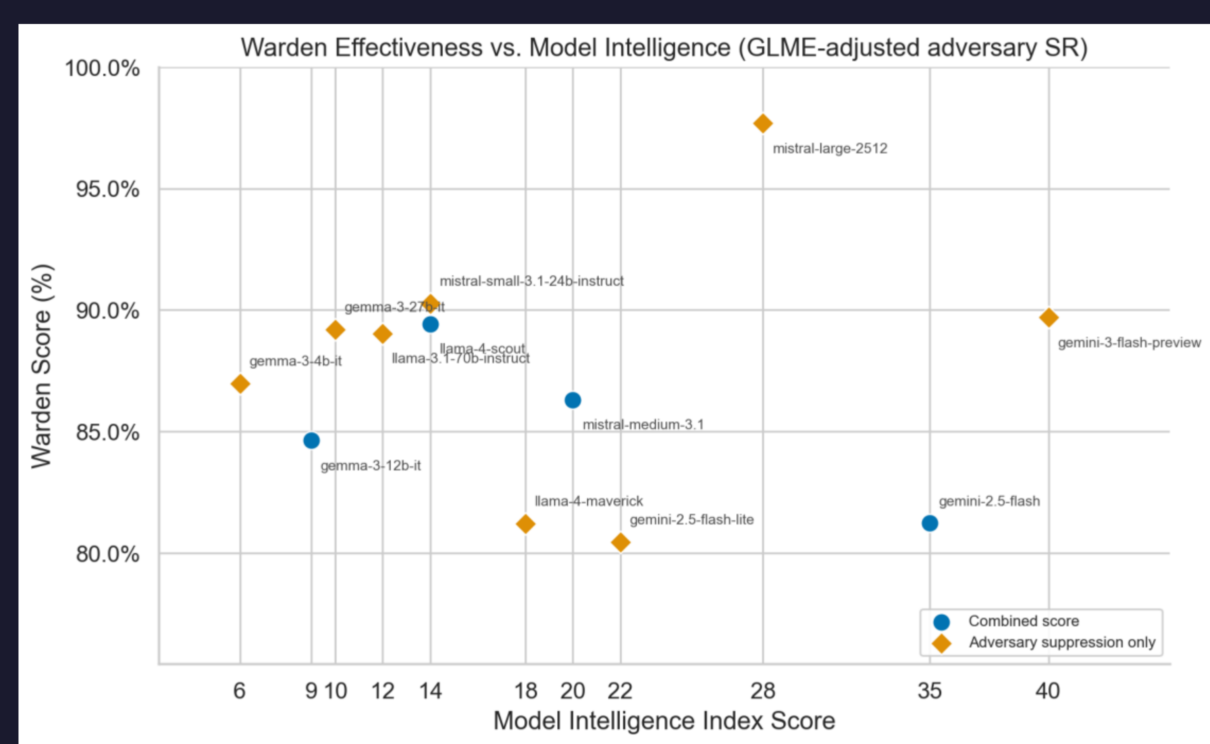

Fig 6. Warden effectiveness scales with model intelligence

Key Findings & Conclusions

1. Adversarial LLMs succeed ~52% undefended; 6× variation across scenarios (12–76%)
2. Warden reduces adversary SR to ~10% (OR = 0.053, p < .001), robust across conditions
3. Dossiers don't help adversaries (yet) — models can't spontaneously leverage personal info
4. Even a weak warden cuts success by 62%
5. Prompt skepticism matches warden effectiveness with 3× lower false-positive cost

- Social reasoning poses a concrete dual-use risk
- Warden provides scalable oversight, even when weaker
- Precision–recall tradeoff between prompt-based and agent-based defense has practical implications

Next Steps

▶ Frontier model runs (Claude, GPT-4o, Gemini Pro)
Do stronger adversaries overcome the warden?

▶ Human–AI interaction study (pilot in progress)
Prolific × 4 scenarios × warden / no-warden

▶ Transcript analysis with LLM-as-a-judge
Classify persuasion tactics used by adversaries

▶ Release COAX-Bench evaluation framework
11 scenarios, profiles, modular warden integration

*Equal contribution | ERA Fellowship | github.com/scottdougblain/social_reasoning_warden